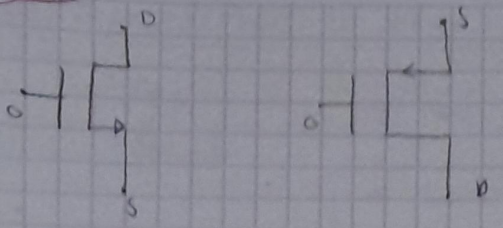
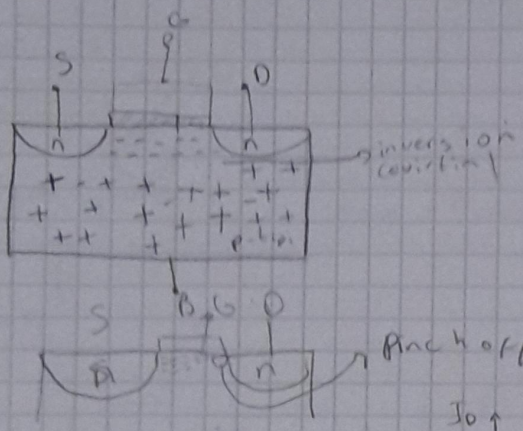
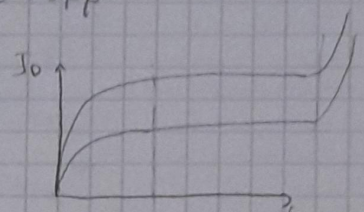
11. hafta

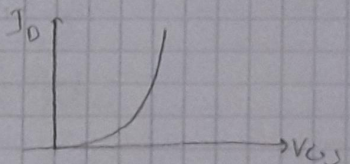


Nmos

Pmos — $V_{TH_p} < 0$







Triode

$$I_D = k_n' \frac{W}{L}\left[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2\right]$$

$$= \frac{1}{2} k_n' \frac{W}{L}\left(2(V_{GS} - V_{TH})V_{DS} - V_{DS}^2\right)$$

$V_{DS} \leq V_{GS} - V_{TH}$

Sat

$$I_D = \frac{1}{2} k_n' \frac{W}{L}(V_{GS} - V_{TH})^2$$

$V_{DS} \geq V_{GS} - V_{TH}$

Deep triode

$$R = \frac{1}{\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH})}$$

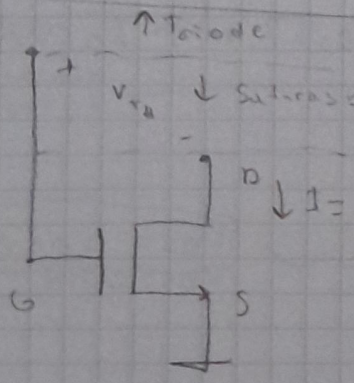
At the edge of saturation

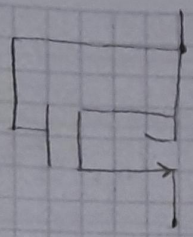
$$V_{DS} = V_{GS} - V_{TH}$$

$$V_D - V_S = V_G - V_S - V_{TH}$$

$$V_{TH} = V_G - V_D$$

$$V_{TH} = V_{GD}$$





Diode connected transistor

Örnek 5.2

0,18 μm process

$L = 0,18\mu m$, $W = 2\mu m$, $C_{ox} = 8,6 fF/\mu m^2$, $\mu_n = 450 cm^2/Vs$, $V_{TH} = 0,5V$

a) $V_{GS} = ?$ $V_{DS} = ?$ at the edge of saturation. $I_D = 100\mu A$

$k'_n = \mu_n C_{ox} = 450 \cdot 10^{-4} \cdot 8,6 \cdot 10^{-15} \cdot 10^{12} A/V^2$

$= 387 \mu A/V^2$

$k_n = k'_n \left(\frac{W}{L}\right) = 387 \left(\frac{2}{0,18}\right) = 4,3 mA/V^2$

$I_D = \frac{1}{2} k_n (V_{ov})^2$

$100 = \frac{1}{2}(4,3) \cdot 10^3 \cdot (V_{ov})^2$

$V_{ov} = 0,22V \rightarrow V_{DS} = V_{GS} - V_{TH} = 0,22$

$V_{GS} = V_{TH} + V_{ov} = 0,5 + 0,22 = 0,72$

b) V_{GS} is kept constant $\rightarrow V_{DS} = ?$ when $I_D = 50\mu A$ azaldı (geriye bak)

Triode bölgesi

$I_D = k_n \left[V_{ov} V_{DS} - \frac{1}{2} V_{DS}^2\right]$

$50 = 4,3 \cdot 10^3 \left[0,22 V_{DS} - \frac{1}{2} V_{DS}^2\right] \rightarrow$ $V_{DS} = 0,06V$ (sınır) / $V_{DS} = 0,39V$ (satura)

$V_{DS} = 0,06V$

c) MOSFET → as a linear amplifier → (cont.)

$V_{DS} = 0,3\ V$

small signal

The change in i_D resulting from $V_{GS} \rightarrow 0,7\ V$ → +0,01 V ⇒ i_D?

→ −0,01 V = i_D?

$V_{GS} = 0,7\ V$, $V_{OV} = 0,2V$ $(V_{t,h} = 0,5\ V)$

$V_{DS} = 0,3\ V$

$$I_D = \frac{1}{2} k_n (V_{OV})^2$$

$$= \frac{1}{2} 4300 \cdot 0,04$$

$$= 86\mu A$$

$V_{GS} = 0,71\ V \Rightarrow V_{OV} = 0,21\ V$

$V_{DS} = 0,3\ V$ →

$$i_D = \frac{1}{2} \cdot 4300 (0,21)^2 = 94,8\mu A$$

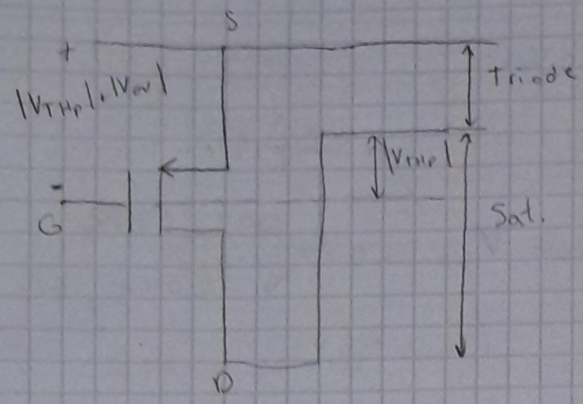
$V_{GS} = 0,69\ V$, $V_{OV} = 0,19\ V$

$$i_D = \frac{1}{2} \cdot 4300 \cdot (0,19)^2 = 77,6\mu A$$

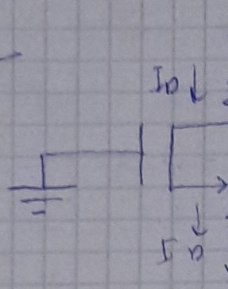
$\Delta V_{GS} = 0,01\ V \rightarrow \Delta_{i_D} = 8,8\ mA$

$\Delta V_{GS} = -0,01\ V \rightarrow \Delta i_D = -8,4mA$

Direkt



Ex. 5.3



$I_D = 0,4\,mA$, $V_{TH} = 0,7V$, $\mu_n C_{ox} = 100\,\mu A/V^2$

$L = 1\,\mu m$, $W = 32\,\mu m$

$(\lambda = 0)$

$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2$

$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{2,5 - 0,5}{0,4} = 5\,k\Omega$

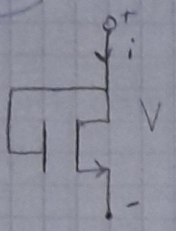
$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2$

$400 = \frac{1}{2} \cdot 100 \cdot \frac{32}{1} V_{ov}^2 \Rightarrow V_{ov} = 0,5V$

$V_{GS} = 0,7 - 0,5 = 1,2V$

$R_S = \frac{-1,2 - (-2,5)}{0,4} = 3,25\,k\Omega$

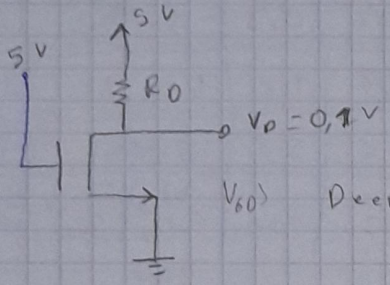
ex: 5.4



→ Da ima ampyrni bilgede

$$i = \frac{1}{2} k_n (v - V_{TH})^2$$

ex: 5.5



$V_{TH} = 1V$, $k_n' \left(\frac{W}{L}\right) = 1 mA/V^2$

→ Effective resistance → $\frac{V_{DS}}{I_{DS}} = ?$

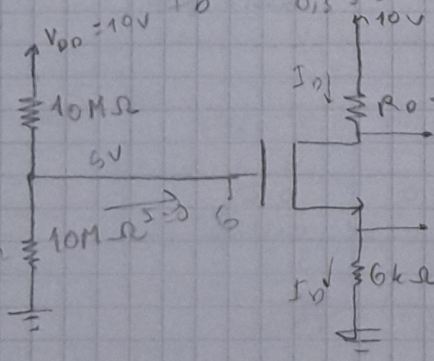
$$I_D = k_n' \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

$$I_D = 1 \cdot \left[(5-1)(0,1) - \frac{1}{2}(0,1)^2 \right]$$

$$= 0,395 mA$$

$$R_{DS} = \frac{V_{DS}}{I_D} = \frac{0,1}{0,395} = 253 \Omega \quad / \quad R_{DS} = \frac{1}{k_n V_{OV}} \Rightarrow \frac{1}{1 \cdot (5-1)} = 250 \Omega$$

ex: 5.6



$V_{TH} = 1V$, $k_n' (W/L) = 1 mA/V^2$ (20)

Determine all voltages and currents

$$V_G = V_{DD} \frac{R_{G2}}{R_{G2} + R_{G1}}$$

$$= 10 \frac{10}{10+10} = 5 V$$

→ Biasing

$I_D = 0,5 mA$

$V_S = 0,5 \cdot 6 = 3V$

$V_G = 5 - 3 = 2V$

$V_D = 10 - 6 \cdot 0,5 = 7 V$

→ V_{TH} ✓

Varsayım 1 sat

$$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_{TH})^2$$

$$I_D = \frac{1}{2} \cdot 1 \cdot (5 - 6I_D - 1)^2$$

$$18 I_D^2 - 25 I_D + 8 = 0$$

$I_{D1} = 0,72$, $I_{D2} = 0,5$

Burası since V_{GS} negatif olur

Appendix C

Comparison of the Mosfet and the BJT

Typical values of CMOS device parameters

	0,8 µm		0,5 µm		0,25 µm		0,18 µm		0,1 µm		65
Parameter	NMOS	PMOS	NMOS	PMOS	NMOS	PMOS	NMOS	PMOS	NMOS	PMO	NMOS
t_{ox}	15	15	9	9	6	6	4	4	2,7	2,7	1,4
C_{ox}	2,3	2,3	3,8	3,8	5,8	5,8	8,6	8,6	12,8	12,8	25
μ	550	250	500	180	460	160	450	100	400	100	270
μC_{ox}	127	58	190	68	267	93	387	86	511	128	540
V_{THO}	0,7	-0,7	0,7	-0,8	0,5	-0,6	0,5	-0,3	0,4	-0,4	0,35
V_{DD}	5	5	3,3	3,3	2,5	2,5	1,8	1,5	1,2	1,2	1,0
$\lvert V_A' \rvert$	25	20	20	10	5	6	5	6	5	6	3
C_{OV}	0,2	0,2	0,4	0,4	0,3	0,3	0,32	0,33	0,36	0,33	0,35

→ NMOS ta hep daha büyüktür çünkü taşıyıcılar daha hareketlidir

Table 6.2

BJT

	High voltage		Low voltage	
Parameter	npn	Lateral pnp	npn	Lateral pnp
A_E (μm^2)	500	500	2	2
I_S (A)	5×10^{-15}	$2 \cdot 10^{-15}$	$6 \cdot 10^{-17}$	$6 \cdot 10^{-15}$
β_F	200	50	100	50
V_A (V)	130	50	32	30
τ_F	0,35 ns	30 ns	10 ps	650 ps

Page 6.7

Example 6.1

a) NMOS → $\frac{W}{L} = 10$, 0,18 μm process

V_{ov}, $V_{GS} = ?$ → $I_D = 100\ \mu A$, ($\lambda = 0$)

b) Table 6.2 → nMOS → low voltage process → $I_D = 100\ \mu A$

$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right) V_{ov}^2$

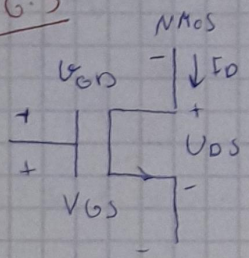
$100\ \mu A = \frac{1}{2} \cdot 387 \cdot 10 \cdot V_{ov}^2 \Rightarrow 0,23\ V = V_{ov}$

$V_{GS} = V_{TH} + V_{ov} = 0,5 + 0,23 = 0,73\ V$

b) $I_C = 100\ \mu A$ $\quad I_C = I_S e^{V_{BE}/V_T}$ → $I_S = 6 \cdot 10^{-17}$

↳ $V_{BE} = 0,025 \times \ln \frac{100 \cdot 10^{-6}}{6 \cdot 10^{-17}} = 0,76\ V$

Table 6.3



Active ① $V_{GS} > V_{TH}$; $V_{TH} \approx 0,3\ V - 0,5\ V$

$V_{GS} = V_{TH} + V_{ov}$

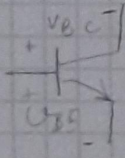
② Pinch off

$V_{GD} < V_{TH}$

$V_{DS} > V_{ov}$, $V_{ov} \approx 0,1\ V \longleftrightarrow 0,3\ V$

$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 \left(1 + \frac{V_{DS}}{V_A}\right)$

$I_G = 0$



$V_{BE} \geq V_{BE\,on}$, $V_{BE} \approx 0,8$, Forward bias (FBJ)

Reverse bias

$V_{BC} < V_{BC\,on}$ / $V_{BC\,on} \approx 0,6\ V$

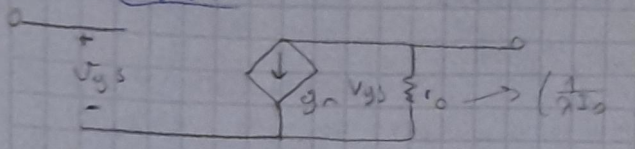
$V_{CE} \geq 0,2\ V$

$I_C = I_S e^{V_{BE}/V_T} \left(1 + \frac{V_{CE}}{V_A}\right)$

$I_B = \frac{I_C}{\beta}$

Small signal

NMOS

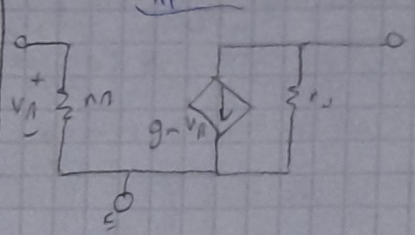


$g_m = \frac{I_D}{V_{ov}/2}$ (Transconductance)

$g_m = \mu_n C_{ox} \frac{W}{L} V_{ov}$

$g_m = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D}$

npn



$g_m = I_C / V_T$

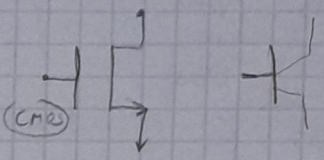
$r_o = V_A / I_C$

$r_\pi = \beta / g_m$

example 6.2

0,25μm process → g_m, r_o, A_0 → intrinsic gain → $A_0 = g_m \cdot r_o$, R_{in}?

npn → 100μA, $L = 0,4\mu m$, $W = 4\mu m$



$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2 \Rightarrow$

$100 = \frac{1}{2} \cdot 2,6 \cdot \frac{4}{0,4} \cdot V_{ov}^2$

MOSFET → $V_{ov} = 0,27 V$

$g_m = \sqrt{2(\mu_n C_{ox})\left(\frac{W}{L}\right) I_D} = 0,73 mA/V$

$R_{in} = \infty$, $r_o = \frac{V_A' L}{I_D} = 20 k\Omega$

$A_0 = g_m r_o = (0,73)(20) = 14,6 V/V$

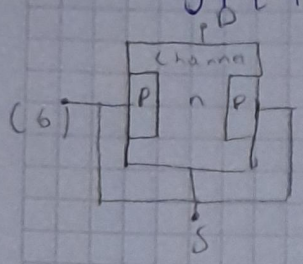
$g_r = \frac{I_C}{V_T} = \frac{0,1}{0,025} = 4 mA/V$

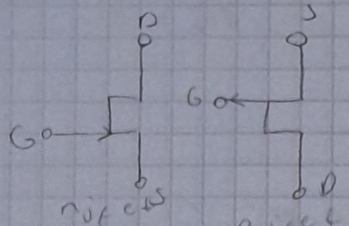
$R_{in} = r_\pi = \frac{\beta}{g_\pi} = \frac{100}{40} = 25 k\Omega$

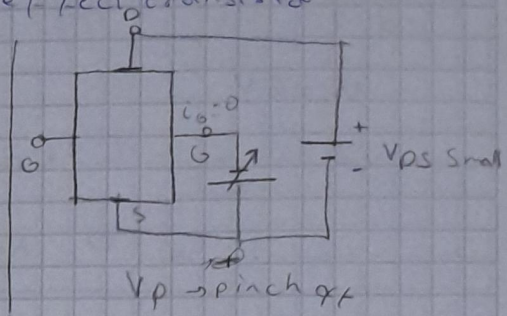
$r_o = \frac{V_A}{I_C} = \frac{35}{0,1} = 350 k\Omega$

$A_0 = g_m r_o = 4 \cdot 350 = 1400 V/V$

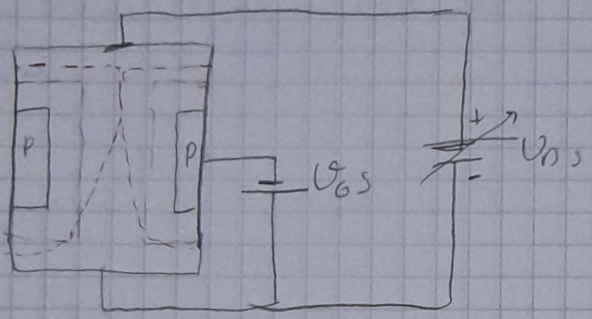
JFET → Junction Field effect transistor







cut off $V_{GS} \leq V_P$, $i_D = 0$



saturation (pinch off) $V_P \leq V_{GS} < 0$

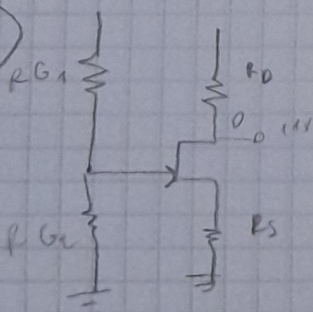
$$V_{DS} \geq V_{GS} - V_P$$

$$i_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 (1 + \lambda V_{DS})$$

Triode $V_P \leq V_{GS} \leq 0$, $V_{DS} \leq V_{GS} - V_P$

$$i_D = I_{DSS}\left[2\left(1 - \frac{V_{GS}}{V_P}\right)\left(\frac{V_{DS}}{-V_P}\right) - \left(\frac{V_{DS}}{V_P}\right)^2\right]$$

(ps 47)



$V_P = -3V$, $I_{DSS} = 9mA$ $\lambda = 0$

$V_G = 5V$, $I_D = 4mA$, $V_D = 11V$

Design for $0.05mA$

$$15 \cdot \frac{R_{G2}}{R_{G1} + R_{G2}} = 5V \Rightarrow \begin{matrix} R_{G1} = x \\ R_{G2} = 2x \end{matrix}$$

$$\frac{15}{3x} = 0.05mA \Rightarrow x = 100k\Omega$$

$R_{G1} = 200k\Omega$
$R_{G2} = 100k\Omega$

$$R_D = \frac{15 - 11}{4mA} = 1k\Omega$$

$V_{DS} \geq V_{GS} - V_P = V_D > V_G - V_P$

↓

$11 > 5 - (-3) = 11 > 7$

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$\sqrt{\frac{I_D}{I_{DSS}}} = \left(1 - \frac{V_{GS}}{V_P}\right) \Rightarrow \frac{2}{3} = 1 + \frac{V_{GS}}{5}$$

$$V_{GS} = 1$$

$V_S = 6\,V$

$$R_S = \frac{6}{4m} = 1{,}5\,k\Omega$$

$i_G = 0$ her zaman

FLM 231 Proje : Son teslim 25.12.2022

